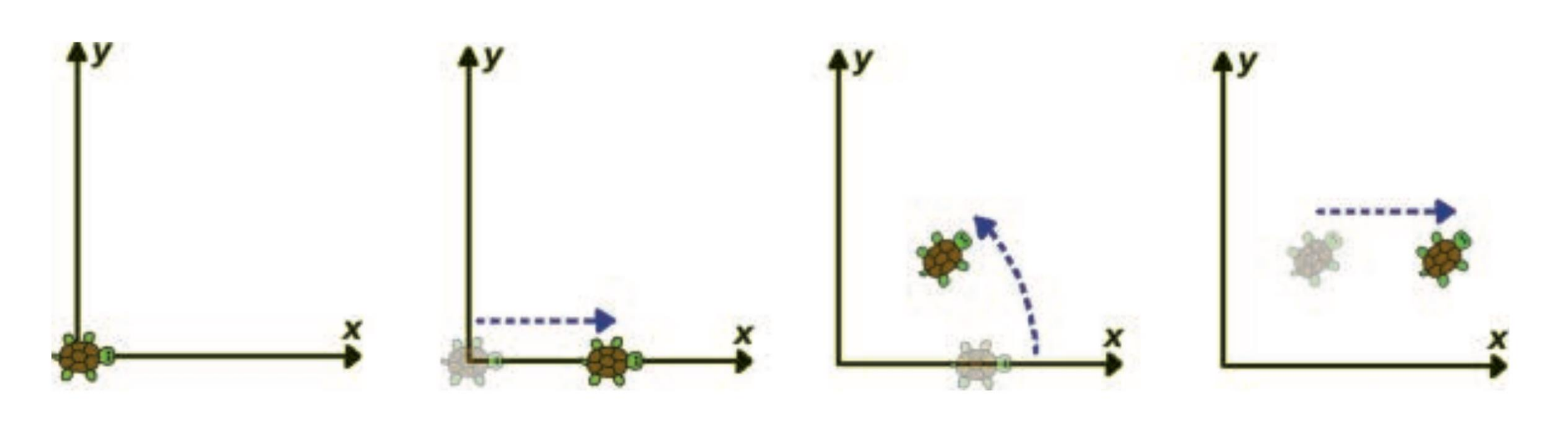
Graphics Programming

Local Transformation

GB. Tony Liang

A Sequence of Global Transformation



A tortoise starts at the origin

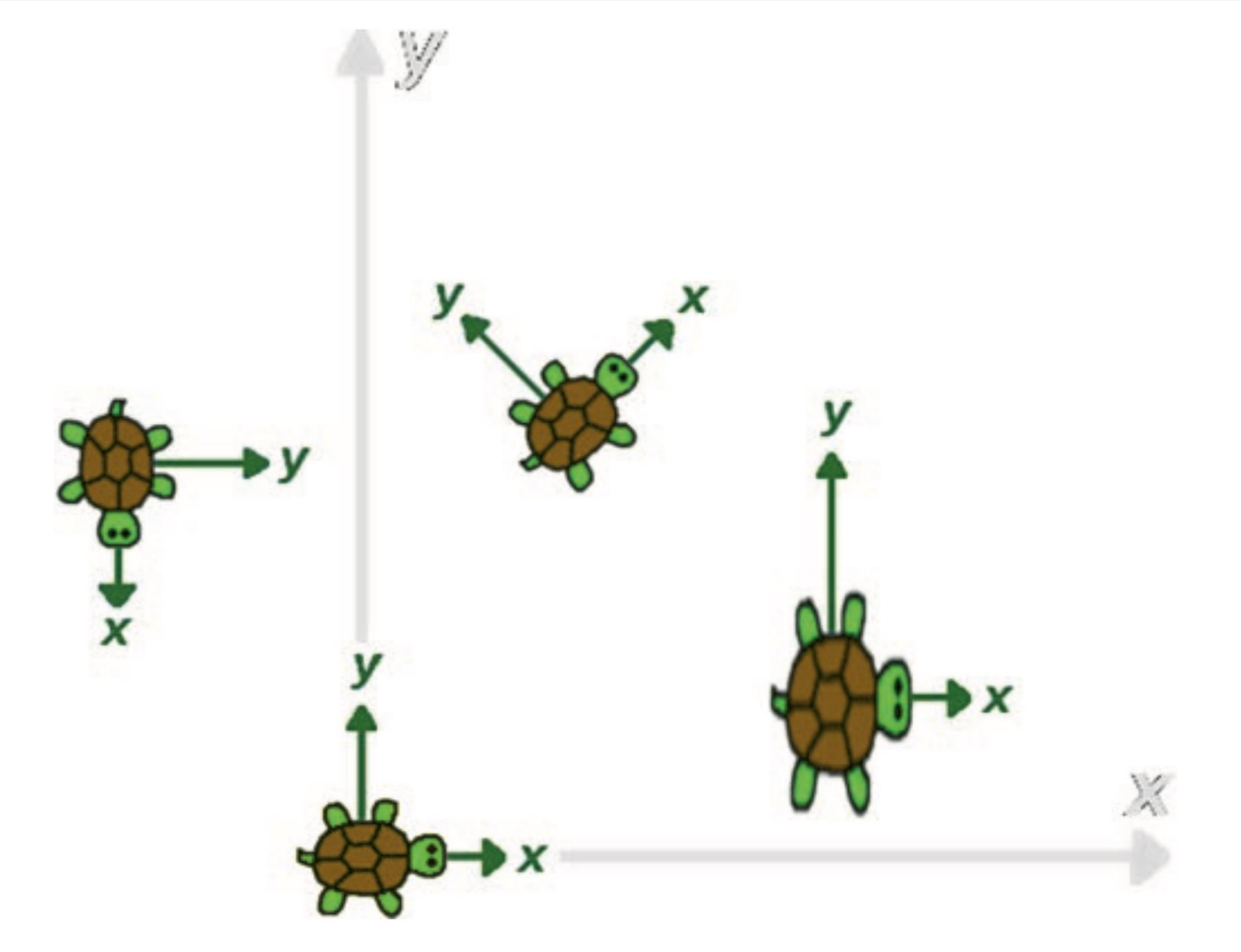
The tortoise moves 2 units to the right

The tortoise rotates 45-degree counter-clockwise

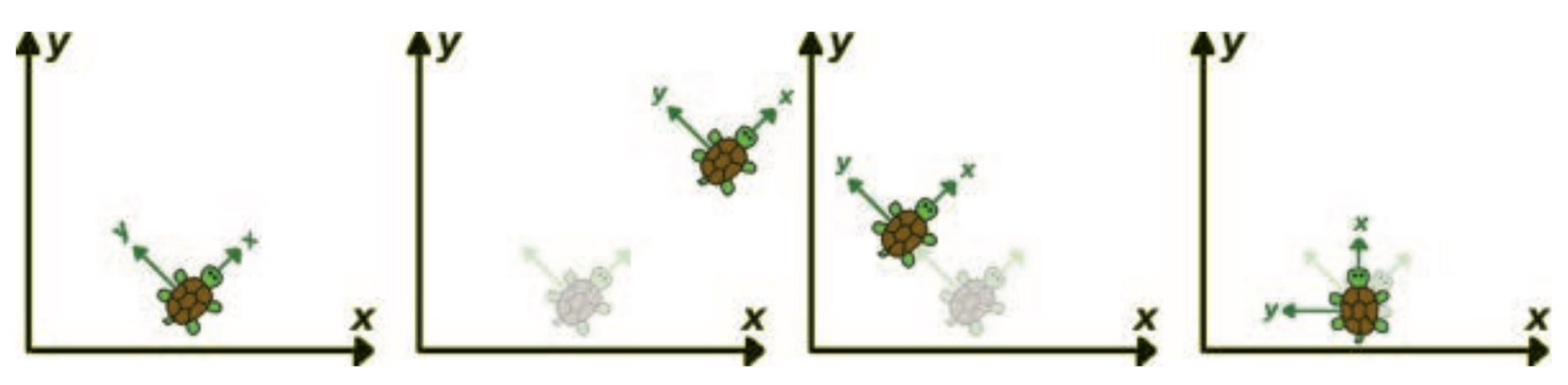
The tortoise moves 2 units to the right

Local Coordinates

- Internal local coordinate system used to define vertices is arbitrary
- Origin point, orientation and scale of the local coordinate axes typically chosen for convenience
  - No reference to global coordinates
- Transformations of an object also apply to local coordinate axes


y
y
x
y
x
y
x
y
x
x

Local Translations



A tortoise starts at a location

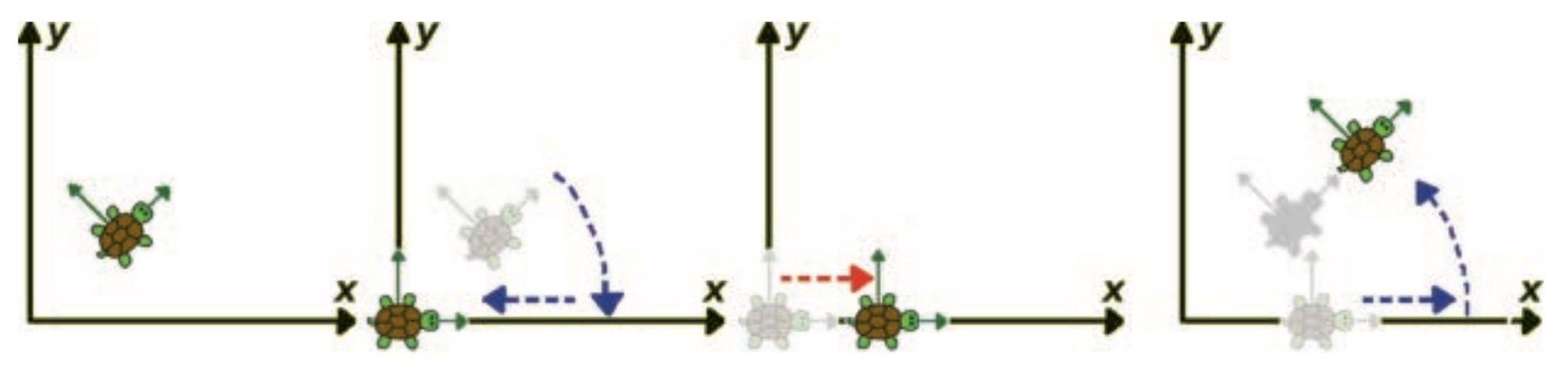
The tortoise moves 2 units towards to its x-axis

The tortoise moves 1 units towards to its y-axis

The tortoise rotates 45-degree counter-clockwise with respect to its origin.

The Model Matrix of an Object

- When transforming a set of points with a matrix
  - Points are multiplied by the matrix in the vertex shader
  - New coordinates are passed alone to the fragment shader
  - New coordinates of the points are **not** permanently stored
- Accumulated transformations applied to object stored as the product of the matrices: ***model matrix***
  - Stores the current location, orientation, and scale of an object



Deriving Local Transformations

- Given model matrix ***M*** and transformation ***T***, apply $\boldsymbol{M^{-1}}$, then ***T***, then ***M***
- Result: $\boldsymbol{M \cdot T \cdot M^{-1} \cdot M}$
  - Left matrix multiplication: transformation
  - Right matrix multiplication: point
  - Equals $\boldsymbol{M \cdot T}$
    - <u>Right multiplication by transformation is a local transformation</u>

Summary

- Apply ***T*** as a **global** transformation
  - Let the new model matrix equal ***T•M***
    - Multiply on LEFT

- Apply ***T*** as a **local** transformation
  - Let the new model matrix equal ***M•T***
    - Multiply on RIGHT